

Aditya Thaker

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EDUCATION

Master of Science in Computer Science, University of Colorado Boulder ; GPA: 3.95 **Aug 2023 – May 2025**

Relevant Coursework: Design and Analysis of Algorithms, Compiler Construction, Distributed Systems, Natural Language Processing, Linux System Administration, Data Center Scale Computing.

Bachelor of Engineering in Computer Engineering, University of Mumbai; GPA: 3.78 **Aug 2019 – May 2023**

TECHNICAL SKILLS

Languages/ Databases: JavaScript, Typescript, Python, Go, C/C++, MongoDB, PostgreSQL, Redis, DynamoDB.

Web Development: Node, React, Next.js, nginx, HTML, CSS, Bootstrap, REST API, gRPC, GraphQL.

Cloud/Big Data: GCP, AWS, GitHub, Spark, PySpark, Kafka, Ansible, Hadoop, Docker, Kubernetes.

AI/ML: TensorFlow, PyTorch, torch-geometric, Langchain, LangGraph, pandas, scikit-learn, numpy.

PROFESSIONAL EXPERIENCE

WalletGyde, Denver, US

May 2025 - Present

AI Engineer

- Built a modular multi-agent chatbot with LangChain.js, OpenAI, and Node.js, improving scalability and reducing orchestration time by 30%
- Integrated Pinecone and Postgres to create a hybrid memory system, cutting user query latency by 25% in demos.
- Adapted a stock dashboard into Walletgyde's personal economy simulator using Next.js and Express.js, achieving 80% feature readiness for Q3 launch.

University of Colorado, Boulder, USA

May 2024 – May 2025

Graduate Researcher

- Built a WebRTC SFU in Go for real-time multi-user conferencing, with support for ICE negotiation, RTP/RTCP handling, and dynamic track renegotiation.
- Improved stream latency and throughput by up to 5% by benchmarking RTP flows and tracing UDP, TCP, and SCTP traffic using eBPF for jitter and packet loss analysis.
- Enhanced observability by logging peer metadata via Cilium eBPF maps and tracing RTP, STUN, DTLS, and QUIC flows at the kernel level for real-time diagnostics.

SimPPL, Mumbai, India

Jun 2022 - Jun 2023

Fullstack Web Developer

- Optimized analytics visualizations, using virtual DOM updates and Canvas rendering, reducing load times by 25%.
- Built an insightful analytics dashboard using d3.js and plotly to track user engagement with news articles, integrating real-time trend analysis.
- Enhanced the REST API with Django, and optimized PostgreSQL schemas, improving response times by 28%.

University of Mumbai, Mumbai, India

Jun 2022 - Jun 2023

Backend Web Developer

- Led end-to-end development of a MERN web app to automate exam duty assignments for 100+ examiners across 8 departments, used Redis to perform CRUD, before persisting on MongoDB.
- Collaborated with stakeholders to define project scope, gather requirements, and ensure the solution aligned with department needs, fostering clear communication and teamwork.
- Configured and deployed application, setup payments with Stripe, and automated emails, reducing manual work.

IIT Patna, Patna, India

Nov 2021 – Jun 2022

ML Research Intern

- Designed a cryptocurrency analytics pipeline using Graph ML to detect fraudulent users in PyTorch.
- Devised a graph ML model to develop a phishing user classifier trained on Ethereum data, achieving 98% F-score.
- Conceptualized ego graphs to convert transactions into temporal transactions, accelerating training time by 20%.

PROJECTS

Content Curation News Feed

- Built an LLM-powered news feed that curates personalized content using vector databases for preference storage.
- Optimized feed generation with Kubernetes, Kafka, and real-time embeddings, cutting load times by 45%.
- Integrated observability with Prometheus and Grafana, enabling proactive tuning and improved reliability.

FinSight – AI Powered Stock Analysis

- Built a research assistant that extracts ESG, operational, and financial insights from news and filings using RAG and LLMs for deep contextual analysis.
- Developed real-time sentiment analysis from Twitter, Reddit, and LinkedIn to capture public perception around companies using Perplexity API.